Step 2: Plot the above ordered pairs (x, y) on the graph using suitable scale.

Step 3: Join the points by a free-hand smooth curve this smooth curve is the graph of $y = x^2 + 2x + 5$

Step 4: The solutions of the given quadratic equation are the x coordinates of the intersecting points of the parabola the X axis.

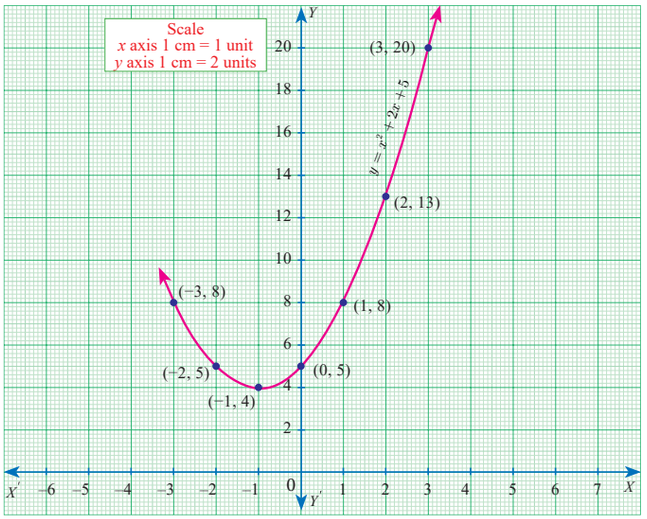


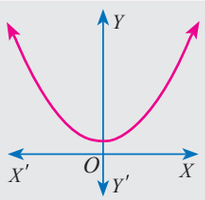
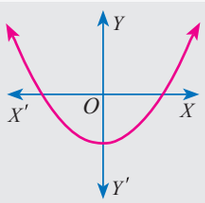
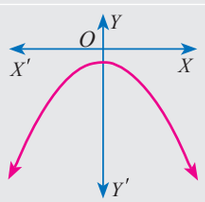
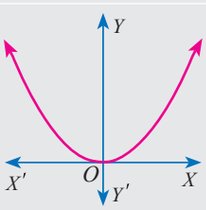
Fig. 3.21

Here, the parabola doesn't intersect or touch the X axis.

So, we conclude that there is no real root for the given quadratic equation.

Progress Check

Connect the graphs to its respective number of points of intersection with X axis and to its corresponding nature of solutions which is given in the following table.

S. No.	Graphs	Number of points of Intersection with X axis	Nature of solutions
1.		2	Real and equal roots
2.		1	No real roots
3.		2	No real roots
4.		0	Real and equal roots

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